

ANJU SAMBASIVAN

Data Analyst • Power BI | SQL | Python | ETL • Christchurch, New Zealand

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New Zealand Permanent Resident • Open to Relocation

PROFESSIONAL SUMMARY

Data and Business Intelligence Analyst with 3+ years of experience transforming complex, multi-source data into analytical insights, BI solutions and automated reporting. Proven experience delivering end-to-end data solutions, analysed data across 2,395 SA2 areas and 850,000+ Police victimisation records, developed automated reporting workflows, and applied predictive modelling to real-world analytical problems. Proficient in Power BI, DAX, SQL, Python and ETL, with experience in data modelling and platforms including Snowflake, Databricks and Microsoft Azure. Master of Applied Data Science graduate from the University of Canterbury and Microsoft Certified: Azure Fundamentals (AZ-900). Open to opportunities where I can apply my data and analytics expertise to solve real-world problems and support data-driven decision-making.

TECHNICAL SKILLS

Business Intelligence & Analytics: Power BI, DAX, Power Query, Data Visualisation, Statistical Analysis

SQL & Databases: Structured Query Language, CTEs, Window Functions, Views, SQL Server, PostgreSQL, SQLite

Data Engineering & Modelling: ETL Pipelines, SSIS, Data Warehousing, Dimensional Modelling, Data Integration

Programming & Data Science: Python (Pandas, NumPy, Matplotlib, Scikit-learn, GeoPandas), R

Cloud & Data Platforms: Microsoft Azure, Databricks, Apache Spark, PySpark

Application Development: REST APIs, React, TypeScript, Node.js, Office.js

Development Tools: Git, GitHub, Azure DevOps, Jupyter, VS Code

AI & LLM: Retrieval-Augmented Generation (RAG), LLM Integration

Currently Upskilling: Microsoft Power BI Data Analyst (PL-300), Azure Data Engineering

PROFESSIONAL EXPERIENCE

Data Scientist • Public Health and Forensic Science

Jan 2026 - Jul 2026

Christchurch, New Zealand

AI-Powered Microsoft Word Reporting Application

Technologies: React • TypeScript • Office.js • Node.js • Express • PostgreSQL • REST APIs • LLMs

- Engineered a Microsoft Word Add-in using React, TypeScript and Office.js to generate and refresh data-driven statistics, tables, charts, maps and analytical content directly within Word.
- Architected reusable frontend components and connected the application to PostgreSQL through Node.js/Express REST APIs, enabling structured retrieval and rendering of analytical data.
- Implemented Office.js content controls and refresh logic to dynamically replace report components when underlying data or analytical outputs changed.
- Integrated LLM-based workflows for summaries, captions and analytical narratives, with researcher validation and editing retained before final report use.
- Validated and debugged data flows across the frontend, REST API, PostgreSQL and Microsoft Word layers to support consistent data retrieval and report rendering.

Data Science Research Intern • Public Health and Forensic Science

Jul 2025 - Jan 2026

Christchurch, New Zealand

Community Health & Wellbeing Analytics

Technologies: Python • Pandas • NumPy • GeoPandas • ETL • Spatial Analysis • Matplotlib • Folium

- Developed Python ETL and geospatial pipelines integrating Census, Police, environmental and air-quality datasets across 2,395 SA2 areas and 850,000+ Police victimisation records.
- Derived population measures, crime rates, wellbeing indicators and weighted composite measures, transforming heterogeneous source data into standardised analytical variables.
- Implemented population-weighted SA2-to-Territorial Authority aggregation and geospatial processing using Pandas and GeoPandas to support analysis at multiple geographic levels.
- Calculated a 2023 victimisation rate of 95.7 per 1,000 people in Christchurch City vs 61.4 in Far North District and identified household crowding of 9.55% in Far North vs 4.61% in Christchurch.

- Automated analytical maps, charts and summary tables using Matplotlib and Folium and designed a metadata-driven framework supporting reusable processing across datasets, indicators, geographic areas and years.
- Produced Community Health & Wellbeing Profiles for research and reporting and presented the analytical methodology and reporting framework at eResearch NZ 2026 in Auckland.

Data Analyst Intern • MVP Studio

Mar 2024 - Jul 2024

New Zealand

NSW Property Analytics - Business Intelligence Solution

Technologies: SQL Server • SSIS • Power BI • DAX • ETL • Data Warehousing • Dimensional Modelling

- Designed a dimensional SQL Server data warehouse integrating property, location and school data from CSV and Excel sources for downstream BI analysis.
- Implemented SSIS ETL pipelines to extract, clean, transform and load source data into structured fact and dimension tables, establishing a repeatable analytical data flow.
- Authored SQL views, transformation queries and validation queries to prepare reporting datasets and verify data completeness and consistency across ETL stages.
- Modelled a Power BI semantic layer with table relationships and DAX measures, enabling analysis by city, suburb, postcode and property-value category.
- Identified Northbridge at \$3.21M as the highest median property value in the analysed dataset compared with approximately \$95K in Broken Hill, highlighting substantial geographic variation.
- Quantified property-value segmentation: 82.91% of analysed records at \$0-\$750K, 14.77% at \$750K-\$1.5M and 2.20% at \$1.5M-\$2.5M.
- Delivered interactive Power BI dashboards and geographic visualisations, validating outputs across SSIS, SQL Server and Power BI to maintain consistency between source data and reported metrics.

Associate Data Analyst • Brain Script Analytica

Feb 2019 - Aug 2021

India

Electricity Load Forecasting - Machine Learning Analysis

Technologies: Python • Pandas • Scikit-learn • XGBoost • TensorFlow • Keras • Machine Learning

- Processed and analysed approximately 3,900 historical electricity load observations in Python to develop and evaluate predictive approaches for electricity-demand forecasting.
- Applied data preprocessing and normalisation techniques, comparing unnormalised, Min-Max and Z-score transformations across six forecasting approaches.
- Trained and compared Linear Regression, SVR, Decision Tree, Random Forest, XGBoost and ANN models using regression, tree-based, ensemble and neural-network techniques.
- Benchmarked model performance using MAE, MSE, RMSE and R², identifying XGBoost with Min-Max normalisation at 84.53% reported performance as the strongest configuration, compared with 84.01% for Linear Regression and 84.43% for the strongest ANN configuration.
- Proposed the strongest-performing model as an approach for predicting substation incomer load from feeder measurements, with potential applications in load monitoring, demand estimation, and identifying unusual load patterns.
- Communicated model comparisons and forecasting findings through analytical visualisations, research outputs and conference presentations on electricity load forecasting.

EDUCATION

Master of Applied Data Science • University of Canterbury | Christchurch, New Zealand | **GPA: 7.50/9.00**

Master of Technology • Computer and Information Sciences • APJ Abdul Kalam Technological University | India | **GPA: 9.63/10.00**

Bachelor of Technology • Computer Science • Mahatma Gandhi University | India | **GPA: 7/9**

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900) - Microsoft
- Databricks Certified Data Engineer Associate - Ultimate Prep - Udemy
- Data Analysis | SQL, Tableau, Power BI & Excel | Real Projects - Udemy
- Docker CI/CD: 40 Practical Demos for DevOps Learners - Udemy
- Microsoft Certified: Power BI Data Analyst Associate (PL-300) - In Progress